

Chapter 10a: Set-Valued Mappings

Applications of Fixed Point Theorems for Multifunctions to Integral Inclusions

Pathak, H.K.

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What is a Set-Valued Mapping?

Definition (Set-Valued Mapping)

A **multivalued mapping** (or **set-valued map**) is a function

$$F : X \rightarrow \mathcal{P}(Y)$$

where $\mathcal{P}(Y)$ denotes the *power set* of Y (the collection of all subsets of Y).

For each point $x \in X$, we have a **nonempty subset** $F(x) \subseteq Y$.

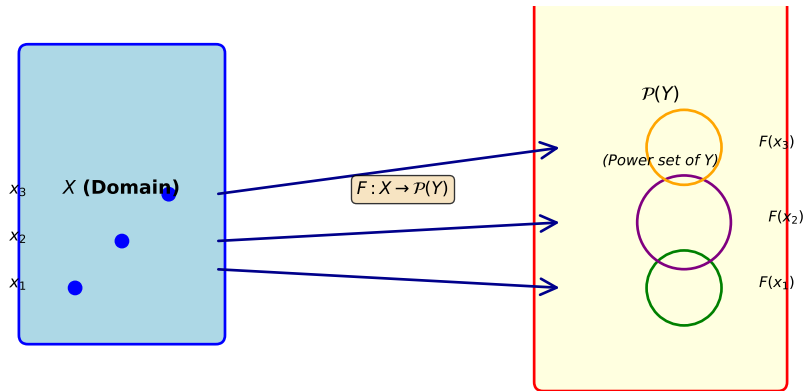
Key difference from single-valued mappings:

- Single-valued: $f : X \rightarrow Y$ maps each x to a single point $f(x)$
- Set-valued: $F : X \rightarrow \mathcal{P}(Y)$ maps each x to a *set* $F(x)$

Applications:

- Differential inclusions and optimal control
- Variational inequalities
- Game theory and economics
- Integral inclusions

Visual Representation of Set-Valued Mappings



Types of Set-Valued Mappings

Definition (Upper Semicontinuity)

A set-valued map $F : X \rightarrow \mathcal{P}(Y)$ is **upper semicontinuous** (u.s.c.) at x_0 if for every open set V with $F(x_0) \subseteq V$, there exists an open neighborhood U of x_0 such that $F(x) \subseteq V$ for all $x \in U$.

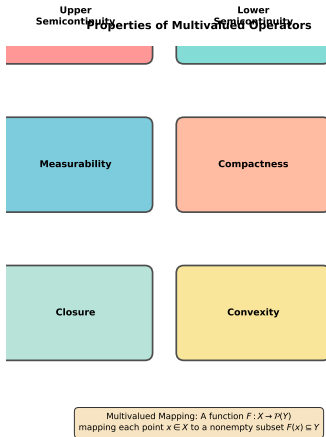
Definition (Lower Semicontinuity)

F is **lower semicontinuous** (l.s.c.) at x_0 if for every open set V with $F(x_0) \cap V \neq \emptyset$, there exists an open neighborhood U of x_0 such that $F(x) \cap V \neq \emptyset$ for all $x \in U$.

Other important properties:

- **Compactness:** $F(x)$ is a compact set for each $x \in X$
- **Convexity:** $F(x)$ is a convex set for each $x \in X$
- **Closedness:** $F(x)$ is closed for each $x \in X$
- **Measurability:** For suitable function spaces

Properties of Multivalued Operators

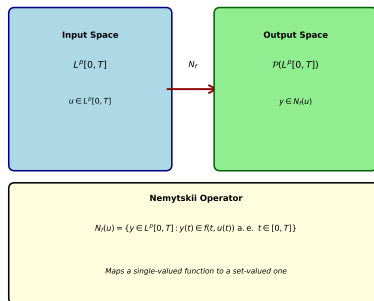


Nemytskii Operators

Definition (Nemytskii Operator)

Given a set-valued mapping $f : [0, T] \times \mathbb{R} \rightarrow \mathcal{P}(\mathbb{R})$, the **Nemytskii operator** is defined as:

$$N_f(u) = \{y \in L^p[0, T] : y(t) \in f(t, u(t)) \text{ a.e. } t \in [0, T]\}$$



Significance: Nemytskii operators convert function spaces of single-valued mappings into function

Integral Inclusions: Basic Framework

Definition (Fredholm Integral Inclusion)

A **Fredholm integral inclusion** is of the form:

$$y(t) \in \int_0^T k(t,s)A(s,y(s)) ds \quad \text{a.e. } t \in [0, T]$$

where:

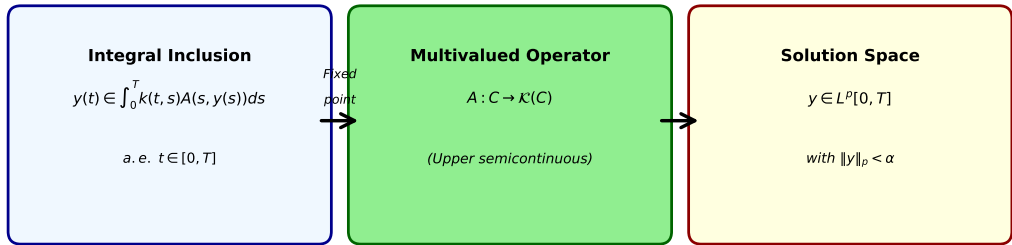
- $k(t,s)$ is a kernel function
- $A(t,y)$ is a set-valued map
- The integral is interpreted in the sense of set-valued analysis

Reduction to Operator Equation:

$$y \in A_K(y) \quad \text{for some } y \in L^p[0, T]$$

This is a **fixed point problem** for the multivalued operator A_K .

Structure of Integral Inclusions



Key Properties:

- Upper semicontinuous mapping
 - Compact-valued
- Fredholm integral inclusion

Fredholm Integral Inclusion with General Setup

Consider the problem:

$$\begin{cases} y(t) \in a(t)g(s, u(s)) + \tilde{f}(s, u(s), u'(s)) & \text{a.e. } s \in [0, T] \\ \sup_{s \in [0, T]} |u'(s)| \leq \lambda \end{cases}$$

Theorem (Theorem 10.1 - Existence of Continuous Solutions)

Let (X, A, μ) be a finite measure space and X a Banach space. Suppose:

- ① $y(s) \in a(s)g(s, u(s)) + \tilde{f}(s, u(s), u'(s))$ for measurable $a, g, \tilde{f}$
- ② $N_f : L_p[0, T] \rightarrow 2^{L_p[0, T]}$ is the generalized Nemytskii operator
- ③ Define $Ay = \{y \in L^p[0, T] : y(t) \in \int_0^T k(t, s)y(s) ds\}$

Then the operator $A : C \rightarrow 2^C$ is **upper semicontinuous and compact**.

Key Properties for Existence

For the integral inclusion to have a solution, we need:

- ① **Upper Semicontinuity** of A :

$A : C \rightarrow \mathcal{K}(C)$ is u.s.c.

where $\mathcal{K}(C)$ is the collection of nonempty compact subsets of C .

- ② **Compactness**: The operator maps to compact-valued sets.

- ③ **Growth Bounds**: For $x \in C \cap \partial\Omega_\alpha$:

$$\|y\|_p < \|x\|_p \quad \text{for } y \in Ax$$

$$\|y\|_p > \|x\|_p \quad \text{for } x \in C \cap \partial\Omega_\beta$$

These ensure by **Theorem 5.167** (multivalued fixed point theorem) that A has a fixed point.

Example: $C[0,T]$ Solutions

Definition ($C[0,T]$)

Problem] Consider the Fredholm integral inclusion:

$$y(t) \in \int_0^T k(t,s)[a(t,s)g(t,y(s)) + \tilde{f}(t,s,y(s))] ds \quad \text{a.e. } t \in [0, T]$$

with finite derivative at each $s \in [0, T]$.

Theorem (Theorem 10.2)

Let $1 \leq p < \infty$ and q be conjugate to p . Assume:

- ① For each $t \in [0, T]$, the map $s \mapsto k(t, s)$ is measurable
- ② $\tilde{f} : [0, T]^2 \times \mathbb{R} \rightarrow \mathcal{K}(\mathbb{R})$ is measurable
- ③ Growth and Lipschitz conditions are satisfied

Then there exists a nonnegative solution $y \in C[0, T]$ to the integral inclusion.

Numerical Example: Integral Inclusion Solution

```
import numpy as np
from scipy.integrate import odeint
import matplotlib.pyplot as plt

# Parameters
T = 1.0
t = np.linspace(0, T, 100)

# Kernel function
def kernel(t, s):
    return np.exp(-(t - s)**2)

# Set-valued mapping (example)
def f(t, y):
    # Returns interval [a(t,y), b(t,y)]
    a = -0.1 * y - 0.5
    b = 0.1 * y + 0.5
    return a, b

# Compute integral inclusion solution
y_solution = np.zeros_like(t)
for i, t_i in enumerate(t):
    integral = 0.0
    for j, s_j in enumerate(t):
        k_val = kernel(t_i, s_j)
        # Take midpoint of f(s_j, y[j])
        f_val = np.mean(f(s_j, y_solution[j]))
```

Hammerstein Integral Inclusions

Definition (Hammerstein Type Problem)

The **nonconvex Hammerstein integral inclusion** is:

$$x(t) = \lambda(t) + \int_0^T k(t, s) F(s, V(x)(s)) ds$$

where:

- $\lambda : [0, T] \rightarrow X$ is a continuous function
- $F : I \times X \rightarrow \mathcal{P}(X)$ is a set-valued map
- $V : C(I, X) \rightarrow C(I, X)$ is a single-valued operator
- $k(t, s)$ is the kernel function

Applications:

- Control systems governed by parabolic PDEs
- Differential inclusions in Banach spaces
- Existence of mild solutions to evolution equations

Semigroup Framework

Definition (C-Semigroup)

A family $\{G(t) : t \in \mathbb{R}^+\}$ of bounded linear operators from X into E is a C_0 -**semigroup** (linear semigroup of class C_0) on X if:

- 1 $G(0) = I$ (identity operator)
- 2 $G(t+s) = G(t)G(s)$ for all $t, s \geq 0$ (semigroup property)
- 3 $G(\cdot)$ is strongly continuous in $t \in \mathbb{R}^+$
- 4 $\|G(t)\| \leq Me^{\omega t}$ for some $M > 0$ and $\omega \in \mathbb{R}$

Significance: Semigroups provide a framework for studying evolution equations and their solutions in infinite-dimensional Banach spaces.

Example: Parabolic PDE as Hammerstein Inclusion

Example

Consider the parabolic system:

$$\begin{cases} x'(t) = Ax(t) + F(t, x(t)) & t \in [0, T] \\ x(0) = x_0 \end{cases}$$

where A is the infinitesimal generator of a C_0 -semigroup $\{G(t)\}_{t \geq 0}$ and F is a set-valued map. A **mild solution** is:

$$x(t) = G(t)x_0 + \int_0^t G(t-s)f(s) ds$$

where $f(s) \in F(s, x(s))$ a.e.

This is a special case of the Hammerstein integral inclusion with $\lambda(t) = G(t)x_0$ and $V(x)(s) = x(s)$.

Existence for Hammerstein Inclusions

Theorem (Existence for Hammerstein Inclusions)

Assume:

- 1 $F : I \times X \rightarrow \mathcal{P}(X)$ has nonempty closed values
- 2 F is measurable and satisfies an L -Lipschitz condition
- 3 $V : C(I, X) \rightarrow C(I, X)$ has continuity and boundedness properties
- 4 $k(t, s)$ satisfies suitable boundedness conditions

Then there exists a solution $x \in C(I, X)$ to the Hammerstein integral inclusion.

Key Steps:

- 1 Define the solution operator: $Tx(t) = \lambda(t) + \int_0^t k(t, s)F(s, V(x)(s)) ds$
- 2 Verify T is upper semicontinuous and compact
- 3 Apply Theorem 5.167 (multivalued fixed point theorem)
- 4 Obtain solution as fixed point of T

Filippov Type Integral Inclusions

Definition (Filippov Problem)

The **Filippov type integral inclusion** is:

$$u(t) \in F(t, V(x(t))) \quad \text{a.e. } (I := [0, T])$$

where:

- $\lambda(\cdot) : I \rightarrow X$ is a given mapping
- $g(\cdot, \cdot) : I \times X \rightarrow X$ is continuous
- $F(\cdot, \cdot) : I \times X \rightarrow \mathcal{P}(X)$ is set-valued
- $f(\cdot, \cdot, \cdot) : I \times I \times X \rightarrow X$ is continuous
- $V : C(I, X) \rightarrow C(I, X)$ is continuous

Connection to Control Systems: Filippov's theorem addresses differential inclusions arising from control problems:

$$x'(t) \in F(t, x(t)), \quad x(0) = x_0$$

where F is upper semicontinuous with closed, convex values

Main Assumptions for Filippov Theorem

[Hypothesis 10.2] Let $F(\cdot, \cdot) : I \times X \rightarrow \mathcal{P}(X)$ be a set-valued map with nonempty closed values that:

(H) $F(\cdot, \cdot)$ is $\mathcal{L}(I) \otimes \mathcal{B}(X)$ measurable

(H) $\exists L(\cdot) \in L_1(I, \mathbb{R}^+)$ such that $F(t, \cdot)$ is $L(t)$ -Lipschitz:

$$H^+(F(t, x), F(t, y)) \leq L(t)\|x - y\| \quad \forall x, y \in X$$

(H) For $\sigma \in L_1(I, E)$ and $T_\sigma(t)$ the solution set:

$$d(\psi_1, T_\lambda(\sigma_2)) \leq H^+(T_\lambda(\sigma_1), T_\lambda(\sigma_2))$$

(H) $a : I \times I \rightarrow \mathbb{R}$ is continuous satisfying Hölder's continuity condition

Filippov Existence Theorem

Theorem (Theorem 10.3 - Filippov Type Existence)

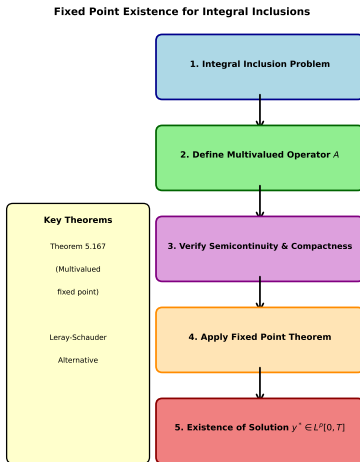
Under Hypothesis 10.2, if $\sup_{t \in I} \|u_0(t)\|$ is bounded and the system parameters satisfy growth and Lipschitz conditions, then the Filippov type integral inclusion admits at least one solution pair (x, u) where:

- $x(\cdot) \in C(I, X)$ is the state trajectory
- $u(\cdot) \in L^1(I, X)$ is the control (selection from F)

Proof Strategy:

- 1 Reformulate as fixed point problem: $x = Tx$ for appropriate operator T
- 2 Prove T is upper semicontinuous with compact, convex values
- 3 Establish a priori bounds on solutions
- 4 Apply Leray-Schauder alternative theorem

Fundamental Results and Tools



Applications of Set-Valued Mappings

1 Control Theory

- Optimal control with constraints
- Bang-bang controls in multivalued systems
- Differential inclusions from control systems

2 Variational Inequalities

- Subdifferentials and monotone operators
- Evolution variational inequalities
- Complementarity problems

3 Partial Differential Equations

- Parabolic and hyperbolic systems
- Semigroup theory for evolution equations
- Integral inclusions with PDE constraints

4 Game Theory and Economics

- Best response correspondences
- Market equilibrium models
- Convex/nonconvex optimization with set-valued objectives

Computational Example: Nonconvex Integral Inclusion

```
import numpy as np
from scipy.optimize import minimize
from scipy.integrate import quad

# Problem: Find solution to nonconvex Hammerstein inclusion
#  $x(t) = \int_0^t k(t-s) f(s, x(s)) ds$ 

def kernel_exp(t, s):
    """Exponential kernel"""
    return np.exp(-2*(t - s))

def f_lower(t, x):
    """Lower bound of set-valued f"""
    return -0.2*x - np.sin(t)

def f_upper(t, x):
    """Upper bound of set-valued f"""
    return 0.2*x + np.sin(t)

def integral_inclusion_residual(x_vals, t_vals):
    """Residual for integral inclusion"""
    residual = 0.0
    for i, t_i in enumerate(t_vals):
        # Compute integral
        integral_val = 0.0
        for j in range(i):
            t_j = t_vals[j]
```

Key Theorems and Results

Theorem (Theorem 5.167 - Multivalued Fixed Point Theorem)

Let K be a nonempty compact convex subset of a Banach space X , and $T : K \rightarrow \mathcal{K}(K)$ be an upper semicontinuous multivalued map with compact convex values. Then T has at least one fixed point in K .

Theorem (Leray-Schauder Alternative)

Let E be a Banach space and $T : E \rightarrow \mathcal{K}(E)$ be an upper semicontinuous multivalued map with compact convex values such that $T(0) = \{0\}$. Then either:

- ① *The fixed point set of T is unbounded, or*
- ② *There exists $\lambda > 0$ and $x \in \partial B_\lambda$ such that $x \in \lambda T(x)$*

Key Concepts Summary

- **Set-Valued Mappings:** Generalize functions to map points to sets
- **Semicontinuity:** Upper/lower semicontinuity replaces ordinary continuity for set-valued maps
- **Nemytskii Operators:** Convert single-valued functions to set-valued operators
- **Integral Inclusions:** Generalize integral equations to include set-valued mappings
- **Fixed Point Theory:** Multivalued fixed point theorems ensure existence of solutions
- **Applications:** Control theory, variational inequalities, PDEs, game theory
- **Semigroups:** Provide framework for solving evolution equations with set-valued terms

Main Results for Integral Inclusions

Problem Type	Main Assumption	Key Theorem
Fredholm Inclusion	USC, Compact	Theorem 10.1
$C[0,T]$ Solutions	Measurability, Growth	Theorem 10.2
Hammerstein Type	Lipschitz, Bounded	Thm 5.167
Filippov Type	Measurable, Lipschitz	Theorem 10.3
General Fixed Point	USC, Compact-convex	Leray-Schauder

Common Framework:

- 1 Reformulate problem as fixed point: $x = T(x)$ or $x \in T(x)$
- 2 Verify properties: continuity, compactness, convexity
- 3 Apply fixed point theorem
- 4 Existence guaranteed

Further Research Directions

- **Uniqueness of Solutions:** When are solutions unique rather than just existent?
- **Regularity Theory:** Can we prove solutions have additional smoothness?
- **Numerical Methods:** How to approximate solutions computationally?
- **Nonconvex-Valued Maps:** What happens without convexity assumptions?
- **Optimal Control:** Can we find optimal selections from set-valued maps?
- **Infinite-Dimensional Spaces:** Extensions to more general Banach spaces?
- **Stability:** How do solutions depend on problem parameters?

References and Further Reading I

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Set-valued mappings and integral inclusions provide a powerful framework for:

- **Modeling** complex systems with uncertainty
- **Solving** existence problems in nonlinear analysis
- **Applying** fixed point theory to practical problems
- **Extending** classical analysis to multivalued settings

The combination of set-valued analysis with fixed point theory creates a rich mathematical framework with wide-ranging applications.

Key Takeaway: Multivalued functions are not just a theoretical curiosity—they are essential for modern mathematics and its applications to control, optimization, and dynamical systems.